Triangle Books with a Page-Rooted Two-Edge Tail: Fractional Tail Convexity

Abstract

Let P_m be a book of $m \geq 1$ triangles on a common spine edge, with one otherwise disjoint path of length two attached to a page vertex. We prove the chromatic-polynomial graphon bound for every P_m and every $p \geq 1/2$. Symmetrization over the page orbit distributes the rooted two-edge-path density $A = T_W d$ with exponent $1/m$. Conditional factorization over the pages and Jensen's inequality reduce the problem to the convex scalar function

$$a^{1/m} \max\{2a - p, 0\}.$$

This yields the sharp target directly from $\int A = \int d^2 \geq p^2$. The case $m = 2$ proves the formerly open NetworkX Atlas graph 123.

1 The family and its target

Let (Ω, μ) be an arbitrary probability space and let $W \colon \Omega^2 \to [0, 1]$ be symmetric and measurable. Put

$$p = \int_{\Omega^2} W(x, y) \, \mathrm{d}\mu(x) \, \mathrm{d}\mu(y), \qquad d(x) = \int_{\Omega} W(x, y) \, \mathrm{d}\mu(y),$$

and define

$$A(x) = (T_W d)(x) = \int_{\Omega} W(x, y) d(y) \, \mathrm{d}\mu(y).$$

The rooted triangle density is

$$\tau(x) = \int_{\Omega^2} W(x, y) W(x, z) W(y, z) \, \mathrm{d}\mu(y) \, \mathrm{d}\mu(z).$$

All integrands used below are bounded, nonnegative, and measurable.

For an integer $m \geq 1$, let P_m have spine vertices a, b, page vertices $z_1, \ldots, z_m$, and tail vertices u, v. Its edge set is

$$ab, \qquad az_i, bz_i \quad (1 \leq i \leq m), \qquad z_1 u, uv.$$

Thus the two-edge tail is rooted at one page, not at a spine endpoint.

After the ordered spine is properly coloured, each page has $x - 2$ choices. The two successive tail vertices each have $x - 1$ choices. Consequently

$$\chi_{P_m}(x) = x(x - 1)^3 (x - 2)^m, \qquad \chi(P_m) = 3. \tag{1}$$

For $q = 1 - p$, direct substitution gives the polynomial target

$$\Phi_{P_m}(p) = q^{m+4} \chi_{P_m}(1/q) = p^3 (2p - 1)^m. \tag{2}$$

The required density interval is therefore $p \in [1/2, 1]$.

2 Fractional tail convexity

The elementary inequality $rs \geq r + s - 1$ for $r, s \in [0, 1]$, applied to $W(x, y), W(x, z)$ and multiplied by $W(y, z)$, gives

$$\tau(x) \geq 2A(x) - p. \tag{3}$$

Since $\tau \geq 0$, we may use the stronger combined statement

$$\tau(x) \geq \max\{2A(x) - p, 0\}. \tag{4}$$

Lemma 2.1 (Fractional tail convexity)**.** *Let $p > 0$ and $0 < \alpha \leq 1$. The function*

$$f_{p,\alpha}(a) = a^\alpha \max\{2a - p, 0\}, \qquad 0 \leq a \leq 1,$$

is convex and nondecreasing.

Proof. The function vanishes on $[0, p/2]$. On $[p/2, 1]$ it equals

$$2a^{\alpha+1} - pa^\alpha.$$

Its first two derivatives on that interval are

$$\begin{aligned} f'_{p,\alpha}(a) &= a^{\alpha-1}\big(2(\alpha + 1)a - p\alpha\big), \\ f''_{p,\alpha}(a) &= \alpha a^{\alpha-2}\big(2(\alpha + 1)a - p(\alpha - 1)\big). \end{aligned}$$

For $a \geq p/2$, the first bracket is at least $p > 0$. The second bracket is positive because $\alpha \leq 1$. At $a = p/2$, the right derivative is $p(p/2)^{\alpha-1} > 0$, whereas the left derivative is zero. Thus the one-sided derivatives increase across the junction, proving both claims. □

Lemma 2.2 (Fractionally weighted rooted triangle)**.** *If $p \in [1/2, 1]$ and $0 < \alpha \leq 1$, then*

$$\int_\Omega A(x)^\alpha \tau(x)\, \mathrm{d}\mu(x) \geq p^{1+2\alpha}(2p - 1). \tag{5}$$

Proof. Symmetry and Fubini give

$$\int_\Omega A(x)\, \mathrm{d}\mu(x) = \int_\Omega d(x)^2\, \mathrm{d}\mu(x) \geq p^2,$$

where the inequality is Jensen. By (4), Lemma 2.1, Jensen's inequality, and monotonicity,

$$\begin{aligned} \int A^\alpha \tau &\geq \int f_{p,\alpha}(A) \\ &\geq f_{p,\alpha}\left(\int A\right) \\ &\geq f_{p,\alpha}(p^2). \end{aligned}$$

The condition $p \geq 1/2$ is exactly what is needed for $p^2 \geq p/2$. Therefore

$$f_{p,\alpha}(p^2) = p^{2\alpha}(2p^2 - p) = p^{1+2\alpha}(2p - 1),$$

which proves (5). □

3 The page-tail theorem

Theorem 3.1 (Page-rooted two-edge tails on triangle books)**.** *Let $m \geq 1$. Every graphon W of edge density $p \in [1/2, 1]$ satisfies*

$$\boxed{t(P_m, W) \geq p^3(2p-1)^m = \Phi_{P_m}(p).}$$

Proof. Integrating the two tail vertices first gives the book integrand multiplied by $A(x_{z_1})$. The m pages form one automorphism orbit. Average the m representations in which the tail is rooted at page z_j and apply the arithmetic–geometric mean inequality. This gives

$$t(P_m, W) \geq \int_{\Omega^{m+2}} W(x,y) \prod_{j=1}^{m} \Big[W(x,z_j) W(y,z_j) A(z_j)^{1/m} \Big] \, \mathrm{d}\mu^{m+2}. \tag{6}$$

For fixed x, y, put

$$R(x,y) = \int_{\Omega} W(x,z) W(y,z) A(z)^{1/m} \, \mathrm{d}\mu(z).$$

The page variables in (6) are independent, so

$$t(P_m, W) \geq \int_{\Omega^2} W(x,y) R(x,y)^m \, \mathrm{d}\mu^2. \tag{7}$$

Here $p \geq 1/2 > 0$. Normalize $W(x,y)\,\mathrm{d}\mu^2/p$ to a probability measure and apply Jensen to the convex function $r \mapsto r^m$:

$$\int W R^m \geq p^{1-m} \left(\int W R \right)^m. \tag{8}$$

Fubini and the definition of τ give the exact identity

$$\int_{\Omega^2} W(x,y) R(x,y) \, \mathrm{d}\mu^2 = \int_{\Omega} A(z)^{1/m} \tau(z) \, \mathrm{d}\mu(z). \tag{9}$$

Apply Lemma 2.2 with $\alpha = 1/m$ and substitute into (7)–(8):

$$\begin{aligned} t(P_m, W) &\geq p^{1-m} \Big[p^{1+2/m} (2p-1) \Big]^m \\ &= p^3 (2p-1)^m. \end{aligned}$$

This is precisely (2). □

At $p = 1/2$, the target vanishes and every displayed step remains valid; no division by $2p-1$ occurs. At $p = 1$, $W = 1$ almost everywhere and both sides equal one. Every balanced complete k-partite graphon, $k \geq 2$, realizes equality: $d = p$, $A = p^2$, and $\tau = p(2p-1)$ almost everywhere, so all symmetrization, factorization, and scalar inequalities are sharp.

4 Atlas 123 and the formalization boundary

Corollary 4.1 (Atlas 123)**.** *NetworkX Atlas graph* 123*, with graph6 code* `EJy?`*, is positive. It has order six, size seven, chromatic number three, and*

$$\chi_{F_{123}}(x) = x(x-1)^3(x-2)^2.$$

Every graphon of edge density $p \in [1/2, 1]$ satisfies

$$t(F_{123}, W) \geq p^3(2p-1)^2 = \Phi_{F_{123}}(p).$$

Proof. Its NetworkX edge list is

$$04,\ 05,\ 12,\ 13,\ 14,\ 23,\ 24.$$

Take spine 12, pages $3, 4$, and tail $4 - 0 - 5$. This is exactly P_2, so Theorem 3.1 applies. □

The proof uses ordinary homomorphism densities on an arbitrary probability space. All uses of Tonelli–Fubini involve bounded nonnegative integrands. For formal verification, the reusable analytic statement is Lemma 2.2 for real $0 < \alpha \leq 1$, built from the positive-part rooted-triangle bound, fractional scalar convexity, and $\int A = \int d^2$. The remaining ingredients are finite orbit averaging, arithmetic–geometric mean, conditional page factorization, and Jensen under the normalized edge measure. The graph-specific module awaiting inspection is `lean/Taeyoung/Examples/Graph123.lean`.

The tail must be rooted at a page vertex of a single triangle book. The theorem does not cover a tail at a spine endpoint (handled separately in `triangle_book_two_edge_tail.tex`), a two-edge tail on K_4 (Atlas 142), or attachments to a general mixed-chordal graph.

The graph, target, fractional exponents, scalar derivatives, and Atlas identification were reconstructed in exact rational arithmetic.

5 What the Lean formalization actually proves

The machine-checked development is `lean/Taeyoung/Methods/PageTail/{Core,Rows}.lean`, ending at `satisfiesLowerBound_123`. It follows this note except at one place, and covers only the scoped case $m = 2$.

The departure: Lemma 2.1 is not formalized. This note proves that $f_{p,\alpha}(a) = a^\alpha \max\{2a - p, 0\}$ is convex and nondecreasing on $[0, 1]$ — by computing f' and f'' on $(p/2, 1]$ and comparing one-sided derivatives at $a = p/2$ — and then applies Jensen at $\int A \geq p^2$ to obtain Lemma 2.2. The formalization never differentiates anything and never mentions `ConvexOn`. At $\alpha = 1/2$ it proves the affine minorant that Jensen would have produced, directly:

$$p^2(2p - 1) + \Big(3p - \tfrac{1}{2}\Big)(a - p^2) \leq \sqrt{a}\, \max\{2a - p, 0\} \qquad (a \geq 0,\ \tfrac{1}{2} \leq p \leq 1), \tag{10}$$

as `PageTail.tangent_sqrt`. Substituting $a = s^2$ with $s = \sqrt{a} \geq 0$ makes both cases polynomial: for $2a \geq p$ the difference of the two sides is

$$2(s - p)^2\Big(s + \tfrac{p}{2} + \tfrac{1}{4}\Big) \geq 0,$$

a double root at $s = p$ times a positive factor, which is exactly the tangency this note obtains by differentiating; and for $2a < p$ the right-hand side is zero while the left-hand side, increasing in a because $3p - \frac{1}{2} > 0$, is at most its value at $a = p/2$, namely $-p\big(p - \frac{1}{2}\big)^2 \leq 0$.

Integrating (10) needs only $\int A \geq p^2$ and the positive-part bound (4), and yields Lemma 2.2 at $\alpha = 1/2$, which is `PageTail.sqrt_weighted_rootedTriangle`. The general α is not formalized; the same substitution $a = s^{1/\alpha}$ would not stay polynomial, so a different certificate would be needed there.

Scope: only $m = 2$ is formalized. Theorem 3.1 is proved here for every $m \geq 1$, by orbit averaging, arithmetic–geometric mean and Jensen for $r \mapsto r^m$. At $m = 2$ each of those three steps collapses to a Cauchy–Schwarz, and the Lean file does exactly that:

$$R(x,y)^2 \leq S(x,y)G(x,y), \qquad \Big(\iint WR\Big)^2 \leq p \iint WR^2,$$

with $R = \int W(x,\cdot)W(y,\cdot)\sqrt{A}$. The first is the weighted Cauchy–Schwarz of `PureChordal/WeightedCauchySchwarz` applied inside the page with weight $W(x,\cdot)W(y,\cdot)$; the second is the same lemma on $\mu \otimes \mu$ with weight W. Nothing is lost for the catalogue: P_1 is the five-vertex triangle with a two-edge tail, already `verified`, and P_3 has seven vertices.

Everything else transcribes directly. The pointwise bound (3) is `Link.rootedTriangle_ge`; (4) is `max_le` against `rootedTriangle_nonneg`; $\int A = \int d^2$ is `BookTail.integral_pathOp`; and (9) is `PageTail.integral_edge_pageWeightOp`, a weight-parametrized clone of the existing `Link.integral_edge_page` — the page operator had to gain an arbitrary vertex weight, since the page carrying the tail weighs A and the Cauchy–Schwarz that splits it weighs $\sqrt{A}$. The final division by p is `le_of_mul_le_mul_left`, so the argument never divides by $2p - 1$.